

A proof of the conjectured recurrence for OEIS A130655

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Abstract

OEIS A130655 carries a conjectured linear recurrence with polynomial coefficients, of order 3, contributed by R. J. Mathar. We prove it. The generating function of the entry is algebraic, and the recurrence is equivalent to the vanishing of an explicit differential operator applied to it: writing $\theta = x \frac{d}{dx}$, the sequence $b(n) = \sum_i p_i(n) a(n-i)$ has generating function $B(x) = \sum_i x^i (p_i(\theta + i)A)(x)$, so the conjecture holds for all $n > d$ exactly when B is a polynomial of degree d . Since A is algebraic over $\mathbb{Q}(x)$, and every simple algebraic extension of $\mathbb{Q}(x)$ is closed under θ , the residual B can be computed in closed form. Here it equals $8x^2 - 6x$, a polynomial of degree 2, which proves the recurrence for every $n > 2$.

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1 The sequence and the conjecture

OEIS A130655 is “Catalan transform of Catalan numbers $C(n+1)$ ”. It has offset 0 and begins

$$a(0), \dots, a(7) = 1, 2, 7, 28, 119, 524, 2363, 10844, \dots$$

The entry gives the generating function

$$A(x) = \sum_{n \geq 0} a(n) x^n = \frac{-\sqrt{1-4x} + \sqrt{2\sqrt{1-4x}-1}}{2x + \sqrt{1-4x}-1}, \quad (1)$$

and it carries the following comment:

Conjecture: $3 \cdot n \cdot (n-2) \cdot (n+2) \cdot a(n) + 4 \cdot (-10 \cdot n^3 + 21 \cdot n^2 + 7 \cdot n - 15) \cdot a(n-1) + 16 \cdot (11 \cdot n^3 - 47 \cdot n^2 + 57 \cdot n - 15) \cdot a(n-2) - 8 \cdot (2 \cdot n - 5) \cdot (4 \cdot n - 9) \cdot (4 \cdot n - 7) \cdot a(n-3) = 0$. - R. J. Mathar, Mar 01 2015

As of the “Last modified” line on the live entry (2017-03-22, revision 19) the statement is still recorded as a conjecture, and no proof appears among the entry’s links, formulas or programs.

Writing the conjectured relation in the form $\sum_{i=0}^3 p_i(n) a(n-i) = 0$, the coefficient polynomials are

$$p_0(n) = 3n(n-2)(n+2), \quad p_1(n) = -4(10n^3 - 21n^2 - 7n + 15), \quad p_2(n) = 16(11n^3 - 47n^2 + 57n - 15)$$

2 Recurrences as differential operators

Let $A(x) = \sum_{n \geq 0} a(n)x^n$ be the generating function of a sequence, and let $\theta = x \frac{d}{dx}$, so that θ acts on x^m as multiplication by m . For a polynomial p we write $p(\theta)$ for the corresponding differential operator, so that $p(\theta)A = \sum_m p(m)a(m)x^m$.

Lemma 1. Let $p_0, \dots, p_r$ be polynomials and put $b(n) = \sum_{i=0}^r p_i(n) a(n-i)$, with the convention $a(m) = 0$ for $m < 0$. Then

$$B(x) := \sum_{n \geq 0} b(n) x^n = \sum_{i=0}^r x^i (p_i(\theta + i) A)(x).$$

Proof. Fix i and substitute $m = n - i$:

$$\sum_{n \geq 0} p_i(n) a(n-i) x^n = \sum_{m \geq 0} p_i(m+i) a(m) x^{m+i} = x^i \sum_{m \geq 0} p_i(m+i) a(m) x^m = x^i (p_i(\theta + i) A)(x),$$

the last step because $p_i(\theta + i)$ multiplies the coefficient of x^m by $p_i(m+i)$. Summing over i gives the claim. $\square$

Corollary 2. With the notation of Lemma 1, the recurrence $\sum_i p_i(n) a(n-i) = 0$ holds for every $n > d$ if and only if B is a polynomial of degree at most d .

Proof. $b(n)$ is the coefficient of x^n in B , so $b(n) = 0$ for all $n > d$ says exactly that B has no terms above x^d . $\square$

This turns the conjecture into a closed-form identity. The point is that it can then be decided exactly, with no truncation: A is algebraic, so B lies in the same field as A , and one only has to recognise it.

Lemma 3. Let $P(x, y) \in \mathbb{Q}(x)[y]$ be squarefree of degree $m \geq 1$ in y , let $K = \mathbb{Q}(x)[y]/(P)$, and write α for the class of y in K . Then K is closed under θ . Explicitly,

$$\alpha' = -\frac{P_x(x, \alpha)}{P_y(x, \alpha)},$$

the quotient taken inside K , and for $u = \sum_{i < m} c_i(x) \alpha^i$ with $c_i \in \mathbb{Q}(x)$,

$$\theta(u) = x \left(\sum_{i < m} c'_i(x) \alpha^i + \alpha' \sum_{i < m} i c_i(x) \alpha^{i-1} \right),$$

reduced modulo P .

Proof. $P(x, \alpha) = 0$ holds identically in x , so differentiating gives $P_x(x, \alpha) + P_y(x, \alpha) \alpha' = 0$. Because P is squarefree, $\gcd(P, P_y) = 1$ in $\mathbb{Q}(x)[y]$, so $P_y(x, \alpha)$ is a unit of K and the displayed quotient is defined there; its inverse is produced by the extended Euclidean algorithm. The formula for $\theta(u)$ is the chain rule, and reduction modulo P returns the value to the basis $1, \alpha, \dots, \alpha^{m-1}$. Every operation stays inside K . $\square$

Consequently, if $A \in K$ then every $p_i(\theta+i)A$ is in K , and so is B . Writing $B = \sum_{i < m} b_i(x) \alpha^i$ with $b_i \in \mathbb{Q}(x)$, the test of Corollary 2 becomes: B is a polynomial precisely when $b_i = 0$ for every $i \geq 1$ and b_0 has constant denominator. Both are decided by exact cancellation of rational functions.

3 The computation for A130655

The generating function (1) is algebraic over $\mathbb{Q}(x)$; its minimal polynomial is

$$P(x, y) = x^2 y^4 + 4xy^3 + 2xy^2 - y^3 + 1,$$

so (1) lies in $K = \mathbb{Q}(x)[y]/(P(x, y))$ and is the root of that polynomial with the right expansion at the origin. Applying Lemma 1 to the coefficient polynomials of Section 1 and reducing in K

by Lemma 3, every component b_i of B with $i \geq 1$ cancels identically and the remaining rational part collapses to

$$B(x) = 8x^2 - 6x,$$

a polynomial of degree 2.

Theorem 4. *The conjectured recurrence*

$$3n(n-2)(n+2)a(n) - 4(10n^3 - 21n^2 - 7n + 15)a(n-1) + 16(11n^3 - 47n^2 + 57n - 15)a(n-2) - 8(2n-5)a(n-3) = 0$$

holds for every $n > 2$.

Proof. Immediate from Corollary 2 and the displayed value of B . □

Remark 5. The polynomial B is not an error term: it records the boundary of the recurrence. For $n \leq 2$ some of the shifted terms $a(n-i)$ fall outside the range of the sequence, and B collects exactly those contributions. Above that range the relation is exact.

4 Verification

Every number above is an output of a script written separately from this note, and two independent checks were run.

First, the generating function was checked against the entry rather than assumed: the Taylor coefficients of (1) were expanded and compared term by term with the DATA section of A130655, agreeing on all 13 terms compared.

Second, the recurrence itself was evaluated directly on the published terms, in exact integer arithmetic and without reference to the generating function: for each n in range the sum $\sum_i p_i(n)a(n-i)$ was formed from the entry's own values and found to vanish, for all 18 values of n from $n = 3$ upward. This is what Corollary 2 predicts from the degree of B , and it is a genuinely different computation from the symbolic one: it never forms B , never differentiates, and uses only integers.

The symbolic step is exact throughout. The residual is obtained by rational-function cancellation in K , not by series truncation, so the identity $B = 8x^2 - 6x$ is an identity of functions and the theorem holds for all $n > 2$ simultaneously.

References

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